Personal Expense Tracker

Major Project Report

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Abstract

The Personal Expense Tracker is a Python-based application developed to help users manage daily expenses effectively. The project allows users to add, view, save, and analyze expenses. Expense data is stored in a CSV file and represented graphically using Matplotlib.

Objectives

Record expenses efficiently. Categorize expenses. Generate reports. Store data permanently. Visualize spending patterns.

Software Requirements

Python 3.x, VS Code/PyCharm, Matplotlib Library.

Modules Used

CSV Module for reading/writing expense data. Matplotlib for charts. Collections for category-wise aggregation.

Methodology

1. User enters expense details. 2. Data is stored in memory. 3. Data is saved to a CSV file. 4. Reports are generated. 5. Charts are displayed using Matplotlib.

Results

Successfully adds expenses, stores data in CSV, generates reports, calculates total expense, displays category-wise analysis, and visualizes expense charts.

Advantages

Easy to use, beginner friendly, lightweight, and supports data persistence.

Limitations

Command-line interface only and designed for a single user.

Future Enhancements

GUI using Tkinter, database integration, budget alerts, and PDF export features.

Conclusion

The Personal Expense Tracker demonstrates Python concepts such as file handling, functions, lists, dictionaries, exception handling, and data visualization.